

Track 40: VLSI & Digital System Design

June 2026

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Department: ECE | MR ROKESH TECHNOLOGY

Skill path: Boolean Logic → Combinational/Sequential Circuits → Verilog HDL → FPGA → ASIC Flow → Timing Analysis

Related: [Track 20 — Embedded Systems](#) | [Department Map](#)

Track Overview

Who this is for: ECE students targeting VLSI design, FPGA development, and digital IC design roles.

Prerequisites: Digital electronics fundamentals; logic gates, flip-flops.

Tools: Xilinx Vivado / Intel Quartus, Verilog/SystemVerilog, ModelSim, FPGA boards (Basys/Nexys), Logisim (intro).

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Digital design awareness	Verilog modules + simulation
1 Month	80–100	Junior FPGA developer	Working FPGA design
3 Months	250–300	Internship-ready	Processor/controller capstone
6 Months	500–600	VLSI entry-level	RTL design portfolio

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Combinational logic, K-maps, multiplexers, adders	Logic design exercises
D3–4	Sequential circuits: FFs, counters, FSMs	FSM design (traffic light)
D5–6	Verilog: modules, always blocks, testbenches	Simulated Verilog designs
D7–8	FPGA synthesis, constraints, pin assignment	FPGA blink + counter
D9–10	Capstone	FPGA project on board

1-Month Foundation

Week	Topics	Project
W1	Advanced Verilog, hierarchical design	ALU design
W2	FSM design, UART/I2C intro (RTL)	Serial communication module
W3	FPGA IP cores, block RAM, FIFOs	Buffer system design
W4	Timing constraints, simulation vs synthesis	Complete FPGA system

3-Month Job Ready

Month	Topics	Projects
M1	RTL design best practices, linting	UART + SPI controller
M2	Pipelining, memory interfaces	Simple processor datapath
M3	ASIC design flow intro, capstone	RISC processor or DSP block capstone

6-Month Professional

Add: SystemVerilog, UVM intro, static timing analysis, low-power design basics, 3 RTL projects, industry coding standards.

Assessment Rubric

Weekly Quiz 20% | RTL labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Simulation must pass before FPGA synthesis. Enforce synchronous design rules (single clock domain initially).